

Agentic AI Engineering

Fundamentals — Full Practice Test (52 Questions)

A full-length self-assessment across all 24 chapters, weighted toward the load-bearing topics. Mixed tiers:
Recall · Application · Analysis.

Why it matters: This is a **handbook self-assessment**, not an official certification exam — there is no accredited exam for this material, so no passing score or blueprint percentages are implied. Use it to find and close gaps.

How to take it

- Answer all 52 questions in one sitting (aim for ~60–75 minutes) without notes.
- Score yourself with the quick-reference answer grid, then read every explanation — especially for misses.
- Revisit the chapters where you missed the most; each question is tagged with its chapter.

Practice Test — Questions

1. The defining characteristic of agentic AI is that: [CH 1 · RECALL]

- A. The user specifies every instruction step
- B. The user specifies the goal; the agent determines how to achieve it
- C. The agent never uses external tools
- D. The model is always fine-tuned per task

2. A rules-based macro fails the moment a web form field moves; an agent re-reads the page and adapts. This illustrates: [CH 1 · APPLICATION]

- A. Automation vs. autonomy
- B. Recall vs. semantic memory
- C. RAG vs. fine-tuning
- D. Monitoring vs. observability

3. The 2017 innovation that unlocked the scaling laws by modeling all words simultaneously was: [CH 2 · RECALL]

- A. The perceptron
- B. The Transformer
- C. The vector database
- D. The context window

4. An LLM produces a confident but fabricated citation. The root cause is that the model: [CH 3 · APPLICATION]

- A. Ran out of tools
- B. Predicts likely text, not verified text
- C. Had its temperature set to 0
- D. Used the wrong embedding model

5. Why is an LLM alone not an agent? [CH 3 · ANALYSIS]

- A. It cannot generate language
- B. It cannot, by itself, perceive external systems, use tools, remember across sessions, or pursue goals
- C. It has no parameters
- D. It cannot be prompted

6. A prompt requests information; a goal requests: [CH 4 · RECALL]

- A. A longer response
- B. An outcome, with implementation left to the agent
- C. A fine-tuned model
- D. A smaller context window

7. In the Agent Loop, which stage changes the state of external systems? [CH 5 · RECALL]

- A. Perceive
- B. Think
- C. Act
- D. Observe

8. An agent writes 'I need the customer's plan tier before answering' before calling a tool. This is:

[CH 5 · APPLICATION]

- A. Chunking
- B. ReAct (explicit reasoning before action)
- C. Memory poisoning
- D. Metadata filtering

9. Removing feedback (Observe) from the loop would primarily eliminate the agent's ability to:

[CH 5 · ANALYSIS]

- A. Generate text
- B. Detect outcomes and adapt when actions fail
- C. Store embeddings
- D. Classify requests

10. Which is a valid stopping condition for a production agent? [CH 5 · APPLICATION]

- A. A maximum number of steps or a satisfied success criterion
- B. A larger model
- C. A hotter temperature
- D. More tool descriptions

11. If the LLM is the brain, tools are the: [CH 6 · RECALL]

- A. Memory
- B. Hands
- C. Eyes only
- D. Context window

12. An agent runs three independent lookups at once to reduce latency. This is: [CH 6 · APPLICATION]

- A. Sequential reasoning
- B. Parallel tool execution
- C. Fine-tuning
- D. Chunking

13. The single biggest lever on correct tool selection is: [CH 6 · ANALYSIS]

- A. Model size
- B. The quality and clarity of tool descriptions
- C. The number of tools
- D. The temperature

14. Context is to memory as: [CH 7 · RECALL]

- A. Notebook is to whiteboard
- B. Whiteboard is to notebook
- C. Vector is to token
- D. API is to protocol

15. A support agent recalls a customer's tickets across separate sessions. This requires: [CH 7 · APPLICATION]

- A. A larger context window
- B. External (persistent) memory with retrieval
- C. Higher temperature
- D. A router

16. Which memory type is bounded by the context window? [CH 7 · ANALYSIS]

- A. Semantic
- B. Episodic
- C. Working
- D. External

17. Which stage converts each document chunk into a vector for storage? [CH 8 · RECALL]

- A. Retrieval
- B. Indexing
- C. Generation
- D. Chunking

18. Choosing too large a chunk size primarily: [CH 8 · ANALYSIS]

- A. Improves retrieval precision
- B. Reduces retrieval precision and raises cost
- C. Eliminates hallucination
- D. Changes the embedding model

19. An HR agent answers a leave question by retrieving the handbook, then state supplements, then checking employment class. This adaptive multi-step retrieval is: [CH 8 · APPLICATION]

- A. Traditional RAG
- B. Agentic RAG
- C. Fine-tuning
- D. Chain of Thought

20. A search for 'reset password' returns a doc titled 'recover your account credentials.' This works because of: [CH 9 · APPLICATION]

- A. Exact keyword matching
- B. Semantic similarity in vector space
- C. A larger model
- D. A stopping condition

21. Mixing embedding models between indexing and querying: [CH 9 · ANALYSIS]

- A. Improves recall
- B. Silently wrecks retrieval quality due to incompatible coordinate systems
- C. Reduces storage
- D. Has no effect

22. HNSW is best described as: [CH 10 · RECALL]

- A. A fine-tuning method
- B. A widely used Approximate Nearest Neighbor graph algorithm
- C. A prompt template
- D. An embedding model

23. Combining semantic similarity with a structured constraint like department = HR is called: [CH 10 · APPLICATION]

- A. Chunking
- B. Metadata filtering
- C. Reflection
- D. Sampling

24. In the MCP architecture, which component exposes external capabilities via a standardized interface? [CH 11 · RECALL]

- A. Host
- B. Client
- C. Server
- D. Token

25. MCP is to AI tool access what ____ was to the web. [CH 11 · ANALYSIS]

- A. HTML
- B. HTTP
- C. SQL
- D. TCP-only routing

26. Which pattern reasons in internal steps but cannot search, execute, or retrieve? [CH 12 · RECALL]

- A. ReAct
- B. Chain of Thought
- C. Tree of Thoughts
- D. Plan-and-Execute

27. Evaluating several competing cloud architectures by exploring options before committing suits: [CH 12 · APPLICATION]

- A. Chain of Thought
- B. Tree of Thoughts
- C. Reflection
- D. Router

28. A production system that combines ReAct with Reflection demonstrates that: [CH 12 · ANALYSIS]

- A. Only one pattern may be used at a time
- B. Reasoning patterns are often combined in production
- C. Reflection replaces ReAct
- D. Patterns are interchangeable with no trade-offs

29. A MAS is best analogized as: [CH 13 · RECALL]

- A. A single employee
- B. An organization of specialists
- C. A vector database
- D. A context window

30. A threat-intel agent alerts an endpoint agent, which alerts a containment agent, with no central manager. This is: [CH 13 · APPLICATION]

- A. Orchestrator–Worker
- B. Peer-to-peer collaboration
- C. Router
- D. State machine

31. A primary cost of adding more agents is: [CH 13 · ANALYSIS]

- A. Lower accuracy always
- B. Communication overhead, higher cost, and latency
- C. Loss of the LLM
- D. Smaller context windows

32. Which framework specializes in knowledge retrieval / RAG? [CH 14 · RECALL]

- A. CrewAI
- B. LlamaIndex
- C. AutoGen
- D. Bedrock Agents

33. The correct order of decisions when adopting a framework is: [CH 14 · ANALYSIS]

- A. Framework first, then architecture
- B. Architecture first, then framework
- C. Model first, then problem
- D. Prompt first, then objective

34. A pipeline routes a request, plans tasks, builds in parallel, self-improves, gets independently reviewed, then a human approves release. This shows: [CH 15 · APPLICATION]

- A. A single pattern dominates
- B. Patterns composed into a production architecture
- C. That architecture doesn't matter
- D. That models replace architecture

35. The most common over-engineering mistake in agentic AI is: [CH 15 · ANALYSIS]

- A. Using a single agent
- B. Reaching for multi-agent complexity before it's justified
- C. Adding a stopping condition
- D. Writing tool descriptions

36. Which pattern makes autonomous workflows predictable and auditable via gated stage transitions? [CH 15 · RECALL]

- A. State Machine
- B. Router
- C. Reflection
- D. Event-Driven

37. Which is NOT one of the three AI-specific additions to the security triad? [CH 16 · RECALL]

- A. Reasoning Integrity
- B. Behavioral Control
- C. Decision Accountability
- D. Availability

38. A documentation agent is given only doc-repository access and no financial-database permissions. This applies: [CH 16 · APPLICATION]

- A. Least privilege
- B. Chunking
- C. Sampling
- D. Temperature control

39. Guardrails primarily make an AI system: [CH 16 · ANALYSIS]

- A. Smarter
- B. More predictable within safe boundaries
- C. Faster at inference
- D. Cheaper to train

40. The through-line across prompt injection, memory poisoning, and retrieval poisoning is that they target: [CH 16 · ANALYSIS]

- A. The network layer
- B. Content the model trusts (prompts, memory, retrieval)
- C. The GPU
- D. The stopping condition

41. Monitoring answers predefined questions; observability additionally: [CH 17 · ANALYSIS]

- A. Investigates unknown problems and explains causes
- B. Only reports uptime
- C. Replaces evaluation
- D. Removes the need for logs

42. Which observability pillar follows one request through its entire lifecycle? [CH 17 · APPLICATION]

- A. Logs
- B. Metrics
- C. Traces
- D. Alerts

43. Which is NOT one of the seven components of a production prompt? [CH 18 · RECALL]

- A. Role
- B. Objective
- C. Embedding dimension
- D. Evaluation criteria

44. The system/developer/user prompt hierarchy functions as a security boundary because it: [CH 18 · ANALYSIS]

- A. Shortens prompts
- B. Prevents a user prompt from overriding organizational policy
- C. Increases temperature
- D. Eliminates the need for retrieval

45. In the Enterprise AI Stack, the layer that performs planning, tools, memory, and orchestration is the: [CH 19 · RECALL]

- A. Intelligence layer
- B. Agent layer
- C. Data layer
- D. Experience layer

46. Standardization and reusable shared services primarily prevent: [CH 19 · ANALYSIS]

- A. Model hallucination
- B. AI silos and fragmentation
- C. Token limits
- D. Slow GPUs

47. AI products should be measured primarily by: [CH 20 · ANALYSIS]

- A. Model size
- B. Business outcomes (e.g., hours saved, MTTR)
- C. Number of parameters
- D. Token count

48. Positioning AI to augment a human's judgment rather than replace it is the: [CH 20 · APPLICATION]

- A. Router model
- B. Copilot model
- C. State machine
- D. Swarm model

- 49. In a multi-agent SOC, which action should always require human approval?** [CH 21 · APPLICATION]
- A. Ranking alerts
 - B. Enriching an investigation with threat intel
 - C. Isolating a production server
 - D. Summarizing evidence
- 50. In quantitative finance, agentic AI is most valuable for:** [CH 22 · ANALYSIS]
- A. Perfectly predicting prices
 - B. Improving decision quality under uncertainty and managing risk
 - C. Removing all human oversight
 - D. Guaranteeing returns
- 51. An attack-detection model loses accuracy as attacker techniques evolve, though its code is unchanged. This is:** [CH 23 · ANALYSIS]
- A. Data drift
 - B. Concept drift
 - C. Behavioral drift
 - D. A syntax error
- 52. The handbook says agentic AI most directly amplifies:** [CH 24 · RECALL]
- A. Physical labor
 - B. Information storage
 - C. Reasoning
 - D. Network bandwidth

Quick-Reference Answer Grid

Score yourself here, then read the detailed explanations that follow.

1. B	2. A	3. B	4. B	5. B	6. B	7. C	8. B	9. B	10. A
11. B	12. B	13. B	14. B	15. B	16. C	17. B	18. B	19. B	20. B
21. B	22. B	23. B	24. C	25. B	26. B	27. B	28. B	29. B	30. B
31. B	32. B	33. B	34. B	35. B	36. A	37. D	38. A	39. B	40. B
41. A	42. C	43. C	44. B	45. B	46. B	47. B	48. B	49. C	50. B
51. B	52. C								

Detailed Explanations

1. Correct: B [CH 1 · RECALL]

Why: Agentic AI pursues objectives autonomously — the user defines what; the agent defines how.

2. Correct: A [CH 1 · APPLICATION]

Why: Automation breaks on the unexpected (no rule exists); autonomy reasons about the new situation and adapts.

3. Correct: B [CH 2 · RECALL]

Why: The Transformer examines relationships among all tokens at once, enabling scaling with size, data, and compute.

4. Correct: B [CH 3 · APPLICATION]

Why: Hallucination arises because the model generates plausible (likely) text rather than retrieving verified facts.

5. Correct: B [CH 3 · ANALYSIS]

Why: Those capabilities are built around the model; the brain alone is not an agent.

6. Correct: B [CH 4 · RECALL]

Why: A goal requests an outcome; the agent owns the implementation details.

7. Correct: C [CH 5 · RECALL]

Why: Act interacts with the outside world (APIs, code, SQL, email) — what separates agents from chatbots.

8. Correct: B [CH 5 · APPLICATION]

Why: ReAct forces explicit reasoning before acting, producing a transparent, auditable trace.

9. Correct: B [CH 5 · ANALYSIS]

Why: Observation closes the feedback loop; without it the agent cannot detect failure or adapt.

10. Correct: A [CH 5 · APPLICATION]

Why: Stopping conditions (max steps, budget, success criteria) bound execution and prevent infinite loops.

11. Correct: B [CH 6 · RECALL]

Why: Tools are the hands that let the reasoning brain act on the world.

12. Correct: B [CH 6 · APPLICATION]

Why: Parallel execution runs independent tasks simultaneously, cutting latency.

13. Correct: B [CH 6 · ANALYSIS]

Why: The model relies entirely on descriptions to choose tools; vague/overlapping ones cause wrong-tool errors.

14. Correct: B [CH 7 · RECALL]

Why: Context is the temporary whiteboard; memory is the persistent notebook.

15. Correct: B [CH 7 · APPLICATION]

Why: Cross-session recall needs persistent external memory, since the context window is erased between sessions.

16. Correct: C [CH 7 · ANALYSIS]

Why: Working memory holds what's actively in use and is limited by the context window.

17. Correct: B [CH 8 · RECALL]

Why: Indexing embeds each chunk and stores it in a vector database.

18. Correct: B [CH 8 · ANALYSIS]

Why: Too-large chunks reduce precision and increase cost; too-small chunks lose context. Chunk size balances both.

19. Correct: B [CH 8 · APPLICATION]

Why: Agentic RAG decides whether/what/when to retrieve, reformulating and combining sources adaptively.

20. Correct: B [CH 9 · APPLICATION]

Why: Embeddings place similar meanings near each other; cosine similarity finds them despite no shared words.

21. Correct: B [CH 9 · ANALYSIS]

Why: Different models produce incompatible vector spaces, so comparisons break down and retrieval quality collapses.

22. Correct: B [CH 10 · RECALL]

Why: HNSW navigates a graph of similar vectors, the most common ANN approach for fast similarity search.

23. Correct: B [CH 10 · APPLICATION]

Why: Metadata filtering pairs vector similarity with structured filters for more precise retrieval.

24. Correct: C [CH 11 · RECALL]

Why: The MCP Server exposes external capabilities in a standardized form; the Host is user-facing and the Client is the communication layer.

25. Correct: B [CH 11 · ANALYSIS]

Why: MCP standardizes agent-to-tool communication the way HTTP standardized web communication.

26. Correct: B [CH 12 · RECALL]

Why: Chain of Thought is entirely internal — no external action — which is why ReAct was introduced.

27. Correct: B [CH 12 · APPLICATION]

Why: Tree of Thoughts explores multiple branches in parallel, keeping the strongest — ideal for comparing options.

28. Correct: B [CH 12 · ANALYSIS]

Why: Production systems frequently combine reasoning patterns to get the strengths of each.

29. Correct: B [CH 13 · RECALL]

Why: A Multi-Agent System is an organization of specialized, cooperating agents — not one generalist.

30. Correct: B [CH 13 · APPLICATION]

Why: Peer-to-peer agents communicate directly and the workflow emerges through collaboration rather than central command.

31. Correct: B [CH 13 · ANALYSIS]

Why: More agents mean more coordination, expense, and potential latency — so specialization must earn its complexity.

32. Correct: B [CH 14 · RECALL]

Why: LlamaIndex specializes in document ingestion, indexing, retrieval, and RAG optimization.

33. Correct: B [CH 14 · ANALYSIS]

Why: Architecture should determine framework selection, not the reverse — choosing framework-first adds needless complexity.

34. Correct: B [CH 15 · APPLICATION]

Why: Real systems layer Router + Planner + specialists + Reflection + Evaluator + HITL; composition is the art.

35. Correct: B [CH 15 · ANALYSIS]

Why: Complexity should be earned; single-agent is right more often than teams expect.

36. Correct: A [CH 15 · RECALL]

Why: State machines model workflows as states with condition-gated transitions, enabling auditability.

37. Correct: D [CH 16 · RECALL]

Why: Availability is part of the traditional CIA triad; the AI-specific additions are Reasoning Integrity, Behavioral Control, and Decision Accountability.

38. Correct: A [CH 16 · APPLICATION]

Why: Least privilege gives each agent only the permissions it needs, limiting blast radius.

39. Correct: B [CH 16 · ANALYSIS]

Why: Guardrails don't add intelligence; they make behavior predictable and prevent catastrophic outcomes.

40. Correct: B [CH 16 · ANALYSIS]

Why: Most AI attacks manipulate trusted content and reasoning rather than the code or infrastructure.

41. Correct: A [CH 17 · ANALYSIS]

Why: Observability explains why unknown behaviors happened, going beyond monitoring's predefined symptom reporting.

42. Correct: C [CH 17 · APPLICATION]

Why: Traces reveal the complete execution path, often most valuable for agents where one decision influences the next.

43. Correct: C [CH 18 · RECALL]

Why: The seven components are role, objective, context, constraints, resources, expected output, and evaluation criteria — embedding dimension is not among them.

44. Correct: B [CH 18 · ANALYSIS]

Why: The hierarchy stops a crafted user request from overriding permanent org policy — the structural defense against prompt injection.

45. Correct: B [CH 19 · RECALL]

Why: The Agent layer turns intelligence into action; the Intelligence layer only performs inference.

46. Correct: B [CH 19 · ANALYSIS]

Why: Shared identity, retrieval, guardrails, and observability prevent fragmented, inconsistent per-team silos.

47. Correct: B [CH 20 · ANALYSIS]

Why: Success is measured by business outcomes, not model size or technical vanity metrics.

48. Correct: B [CH 20 · APPLICATION]

Why: The copilot model has AI support decisions while humans retain judgment — often beating full automation for trust and adoption.

49. Correct: C [CH 21 · APPLICATION]

Why: High-consequence, hard-to-reverse actions (isolating production, revoking executive access) require human authorization — AI recommends, humans authorize.

50. Correct: B [CH 22 · ANALYSIS]

Why: AI supports better, consistent decisions under irreducible uncertainty; risk management matters more than prediction.

51. Correct: B [CH 23 · ANALYSIS]

Why: Concept drift is when the input→output relationship changes; evolving attack techniques make historical indicators lose predictive power.

52. Correct: C [CH 24 · RECALL]

Why: Where prior revolutions amplified labor and information processing, agentic AI amplifies reasoning — intelligence as infrastructure.